

JEE MAIN REVISION MASTER: METALLURGY 2026

0 QUESTIONS 0% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 10 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Metallurgy - Core Application	10 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Metallurgy - Core Application (High Yield — 10 PYQs)

Core Principle: Key problem pattern in Kcet Metallurgy.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "The incorrect statement about Hall -Heroult process is..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "Select the correct statement :..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "For the formation of which compound in Ellingham diagram ΔG° becomes more and more negative with increase in temperature?..." [Ans: A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

The incorrect statement about Hall -Heroult process is

- (A) carbon anode is oxidised to CO and CO₂.
- (B) Na₃AlF₆ helps to decrease the melting point of the electrolyte.
- (C) CaF₂ helps to increase the conductivity of the electrolyte.
- (D) oxidation state of oxygen changes in the overall cell reaction.

Correct Answer: None

Solution Analysis: The incorrect statement about Hall-Heroult process is given in option (d). It is because in this process oxidation state of oxygen remains same in the overall cell reaction. $2\text{Al}_2\text{O}_3 + 3\text{C} \rightarrow 4\text{Al} + 3\text{CO}_2$